Security and Privacy

Lab Worksheets

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— PL 4: Performance Measures for Message Digests, Symmetric and Asymmetric Cryptography

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ASSIGNMENT #1: Performance Benchmarking of Cryptographic Mechanisms

Due date: **April 2, 23:59**

Grading: Assignment #1 is worth **3 points**

TO BE DONE IN **GROUPS OF THREE (MANDATORY)**

[Cryptography hazmat python manual: https://cryptography.io/en/latest/hazmat/primitives/]

In this exercise you should measure the execution times of cryptographic mechanisms based on AES, RSA and SHA when processing files of different sizes, using your own python implementations.

Some notes:

- You should measure the time of cryptographic operations/algorithms only, not including the time for generation of files and others side aspects.
- If you use padding, this may affect the results specially for small file sizes

A. Generate random text files with the following sizes: 8, 64, 512, 4096, 32768, 262144, 2097152.

B. Encrypt and decrypt all these files with AES in Counter Mode, using the code that you wrote previously. Employ a key of **256 bits**. Measure the time it takes to encrypt and decrypt each of the files. To do this, you might want to use the python module `timeit`.

> Make sure to produce statistically significant results (e.g. with standard deviation or confidence intervals). Do results change if you run a fixed algorithm over the same file multiple times? And what if you run an algorithm over multiple randomly generated files of fixed size?

C. Implement the RSA function and its inverse for a key of size **2048 bits** (minimum recommended for RSA).

> One way to encrypt securely with RSA is by computing $Enc(m; r) = (RSA(r), H(0, r) \oplus m_0, \ldots, H(n, r) \oplus m_n)$.
>
> Here r is a uniform, H is a hash function that produces outputs of size ℓ, and n is the number of blocks $\lceil |m|/\ell \rceil$.
>
> Implement this encryption algorithm with $H = SHA256$ and measure its execution time for the files generated in point A.

D. Measure the time for SHA-256 hash generation for the file sizes listed in point A.

E. Prepare a report of your observations, including the following information:

- Code implemented for points b., c., and d. above
- Brief explanation of the main components of the code (the rest should be submitted in a separate compressed file)
- Explain how you generated/obtained the results – must be *statistically significant*. This must include a description of the experimental setup (e.g. computer characteristics, OS, software versions).
- Plots showing: (i) AES-based encryption/decryption times; (ii) RSA-based encryption times; (iii) RSA-based decryption times; and (iv) SHA digests generation times (plots can be combined for easier comparison). In these graphs, the X axis should plot the file sizes in units of bytes, and the Y axis should plot time measurements in units of microseconds (us).
- The report should also analyze and explain the performance results of:
  - Comparison between AES-based encryption and RSA-based encryption.

- Comparison between AES-based encryption and SHA digest generation.
- Comparison between RSA-based encryption and decryption times.

Submit your report in moodle.